

MATÍAS FERNÁNDEZ LAKATOS

Computer Vision Researcher · Senior Neuromorphic Researcher · PhD in Optics

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PROFESSIONAL SUMMARY

Computer vision researcher bridging **physics-based imaging and machine learning**. PhD in Optics focused on quantitative phase imaging, with 10+ years developing image formation, reconstruction, and processing methods validated on experimental optical data. Currently **Senior Neuromorphic Researcher at Gradiant**, researching **event-based vision (DVS sensors)** and **Spiking Neural Networks** for image analysis — where deep knowledge of optics, sensors, and photonics meets modern AI. Also experienced in production deep learning (custom TensorFlow/Keras architectures, MLflow, XAI) and author of 6 peer-reviewed publications, mostly in optics and imaging journals.

PROFESSIONAL EXPERIENCE

Senior Neuromorphic Researcher

Gradiant · Vigo, Spain · *Jul 2026 – present*

- Researching **neuromorphic vision sensors (DVS / event cameras)**: how their event-based, low-throughput output can be integrated into computer vision use cases and applications.
- Investigating **Spiking Neural Network (SNN) architectures** for the analysis of images and time series, targeting massively parallel, **energy-efficient AI on the edge**.
- Exploring how Gradiant's **photonic architectures** can evolve and be linked to neuromorphic computing — a direct bridge with my optics background.
- Investigating **explainable AI (xAI)** techniques for SNN-based vision models.
- Researching SNN-to-FPGA compilation and neuromorphic hardware; exploring applications in autonomous vehicles, manufacturing, and healthcare.
- Reviewing and updating the **state of the art**: new publications, sensors, hardware devices, and software tools.

Research Engineer

Gradiant · Vigo, Spain · *2025 – Jun 2026*

- Designed and implemented unsupervised deep learning models (**β -VAE, Autoencoder, Isolation Forest variants**) for real-time anomaly detection on large-scale data, building custom architectures in **TensorFlow/Keras**.
- Applied **XAI techniques (SHAP)** to make model decisions interpretable for non-technical users.
- Worked within a scalable streaming architecture (**Kafka, Spark, Airflow**); tracked experiments with **MLflow** and managed artifacts with **MinIO**.

Researcher, Lecturer & Teaching Assistant

Faculty of Engineering, Universidad de la República (Udelar) · Montevideo, Uruguay · *2015 – 2024*

- Developed **novel phase retrieval and quantitative phase imaging methods** (Transport of Intensity Equation variants, sinusoidal-mask propagation, single-derivative integration), implemented computationally and validated on experimental optical data.
- Contributed to the development of **multispectral instruments for remote gas monitoring** and built multi-camera acquisition and image-processing pipelines.
- Published **6 peer-reviewed articles** (4 first-author, Scopus), primarily in optics and imaging journals; presented at SPIE, IOP, and Elsevier conferences.
- Taught university-level physics; supervised students in quantitative modeling.

TECHNICAL SKILLS

Computer Vision & Image Processing

[OpenCV](#) · [scikit-image](#) · [PIL/Pillow](#) · [NumPy](#) / [SciPy](#) · Quantitative Phase Imaging · Phase Retrieval · Wavefront Reconstruction · Digital Image Processing · Pattern Recognition · Multi-camera Systems

Neuromorphic Vision (current research)

Event-based Vision (DVS sensors) · Spiking Neural Networks (SNNs) · Synaptic Plasticity · Photonic Neuromorphic Architectures · SNN-to-FPGA Compilation · Edge AI

Deep Learning & ML

[TensorFlow](#) · [Keras](#) · [PyTorch](#) · [scikit-learn](#) · [MLflow](#) · CNNs / Deep Learning for Computer Vision (CS231n) · Autoencoders / β -VAE · Anomaly Detection · Explainable AI (SHAP/XAI) · Reinforcement Learning

Physics & Optics

Coherent Optics · Interferometry · Optical Instrumentation · Signal Processing · Multivariate Statistics · Numerical Optimization

Programming & Tools

[Python](#) · [MATLAB](#) · [R](#) · [SQL](#) · [LaTeX](#) · [Git](#) · [Linux](#) · [Docker](#) (basic)

EDUCATION

- **Ph.D. in Optics (Physics)** — Universidad de la República, Uruguay (2019–2024). Thesis: *Visualization and Characterization of Phase Objects*. Coursework: Deep Learning for Computer Vision (CS231n), Computer Image Processing, Computational Multivariate Statistics, Reinforcement Learning, Coherent Optics.
- **MSc in Big Data Analytics Technologies** — Universidad de Santiago de Compostela, Spain (2024–2025).
- **MSc in Theoretical Physics (QCD)** — Universidad de la República, Uruguay (2016–2018).
- **BSc in Physics** — Universidad de la República, Uruguay (2011–2015).

PUBLICATIONS

First Author (Peer-Reviewed, Scopus Indexed)

1. **[2024 - SPIE]** [Integration with one partial derivative applied to quantitative phase imaging](#)
2. **[2023 - Optics and Lasers in Engineering]** [Common-path quantitative phase imaging by propagation through a sinusoidal intensity mask](#)
3. **[2022 - Optik]** [Phase retrieval by amplifying the prism term of the Transport of Intensity Equation with a sliding step function](#)
4. **[2019 - IJMPA]** [On the contribution of different coupling constants in the infrared regime of Yang-Mills theory](#)

Co-Author

1. **[2025 - Optics & Laser Technology]** [Wavefront reconstruction of discontinuous phase objects using directional derivatives](#)
2. **[2024 - IOP / Measurement Science and Technology]** [Hilbert's and Takeda's single-shot interferometry with a linear-carrier: a comparison](#)

CERTIFICATIONS & ACHIEVEMENTS

- **Neo4j Certified Professional** — Neo4j GraphAcademy, May 2026
- **Master Python for DevOps and CI/CD** — Udemy, 2025

- **Member of the National System of Researchers (SNI)**, Uruguay, 2024 · PhD scholarships (CAP 2022, ANII 2019)

LANGUAGES

Spanish — Native · **English** — Professional working proficiency

Last updated: July 2026